



**PRA
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**PRACTICAL
CENTRE**

CLASS-XII

B

BIOLOGY

CHAPTER # 18

A

CHEMICAL CO-ORDINATION

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1. HYPOTHALAMUS:

Hypothalamus is a part of brain located just below the thalamus and serves as a connecting link between the nervous and endocrine systems. It consists of two groups of neurosecretory cells.

One group of neurosecretory cells produces two types of hormones i.e., **releasing hormones/factors and inhibiting hormones/factors**. Releasing hormones/factors regulate the release of hormones from the anterior lobe of pituitary gland while the inhibiting hormones/factors cause inhibition of hormone from the target glands.

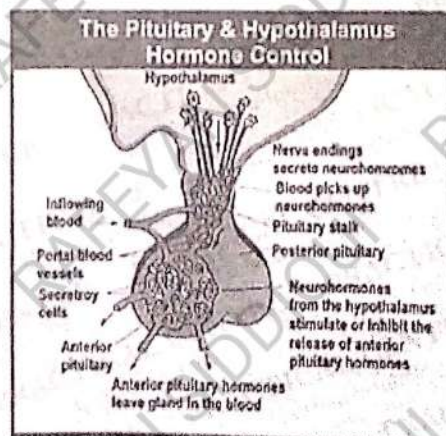
The other group of neuro-secretory cells secretes **Antidiuretic hormone (ADH) and Oxytocin** which are stored in their axon terminals located in the posterior pituitary gland.

RELEASING AND INHIBITING HORMONES OF HYPOTHALAMUS:

S.No.	HORMONE	FUNCTION
1.	Thyrotropin releasing hormone (TRH)	Stimulate the release of thyroid stimulating hormone (TSH)
2.	Corticotropin releasing hormone (CRH)	Stimulate the release of adrenocorticotropin hormone (ACTH)
3.	Gonadotropin releasing hormone (GnRH)	Stimulate the release of LH and FSH.
4.	Growth hormone releasing hormone (GnRH)	Stimulate the release of growth hormone (GH)
5.	Growth hormone inhibiting hormone (GHIH)	Inhibits the secretion of growth hormone.
6.	Prolactin inhibiting hormone (PIH)	Inhibit the secretion of prolactin.
7.	Melanocyte inhibiting hormone (MIH)	Inhibit the secretion of MSH.

2. PITUITARY GLAND: (HYPOPHYSIS)

It is a small pea-sized structure, attached with a short stalk of nerve fibers to the hypothalamus. It was also previously called master gland because of its influence on other endocrine glands. However, pituitary gland is itself under the influence of hypothalamus. It consists of two lobes, the glandular anterior lobe or adenohypophysis and the posterior lobe or fibrous neurohypophysis.



A. ANTERIOR LOBE (ADENOHYPOPHYSIS):

The anterior lobe produces seven essential hormones which stimulate the growth of their target organs and stimulate the production and secretion of hormones from other endocrine glands. A group of hormones secreted by anterior lobe are collectively termed as tropic hormones or tropins. Tropic hormones act on other endocrine glands to stimulate secretion of hormone from these glands.

- i. Adrenocorticotrophic hormone (ACTH): This hormone is protein in nature. It stimulates the release of hormone from adrenal cortex.
- ii. Melanocyte stimulating hormone (MSH): It stimulates the synthesis and dispersion of melanin pigment in the skin.
- iii. Somatotropic hormone (STH) or Growth hormone (GH): This hormone is protein in nature. It stimulates the growth of bones and muscles.
- iv. Prolactin: This hormone stimulates the production of milk from mammary glands.
- v. Thyroid stimulating hormone (TSH): This hormone is Glycoprotein in nature. It stimulates the secretion of thyroxine from the thyroid gland.
- vi. Follicle stimulating hormone (FSH): This hormone is Glycoprotein in nature. It stimulates the formation of follicles in the ovary.
- vii. Luteinizing hormone (LH): This hormone is protein in nature. It stimulates the formation of corpus luteum and ovulation in the ovary.

B. POSTERIOR LOBE (NEUROHYPOPHYSIS):

The hormones, released from posterior lobe of pituitary gland, are stored secretions of hypothalamus. The hormones include:

- i. Antidiuretic hormone (ADH): It increases the reabsorption of water into the blood by kidney and therefore decreases urine output.
- ii. Oxytocin: It stimulates contraction of uterus during labour and release of milk from the mammary glands during breast feeding.

3. THYROID GLAND:

The human thyroid gland is a bi-lobed butterfly-shaped structure, located at the base of neck in front trachea. It secretes three major hormones:

- i. Thyroxin (tetra-iodo thyronine or T₄): It consists of four atoms of iodine and is secreted in greater amount but is less potent than T₃. It influences most of the cells in the body, increasing their metabolic rate. Its functions include increasing consumption of oxygen and heart rate, stimulating breakdown of glucose through synthesis of enzymes.
- ii. Triiodothyronine or T₃: It contains three atoms of iodine. It is released in small amount but is more potent than T₄.
- iii. Calcitonin (Thyrocalcitonin): Raised blood calcium stimulates the production of calcitonin. It causes the deposition of excess calcium in bones.

THYROID DISORDERS:**A: HYPERTHYROIDISM:**

The higher levels of thyroxin in the blood results in Hyperthyroidism. It may produce symptoms such as profuse sweating, weight loss, heat intolerance, irritability and high blood pressure.

B: HYPOTHYROIDISM:

The low levels of thyroxin in the blood results in Hypothyroidism. It may lead to following disorders:

i. CRETINISM:

It occurs due to the deficiency of thyroxin in the early age. Its symptoms include:

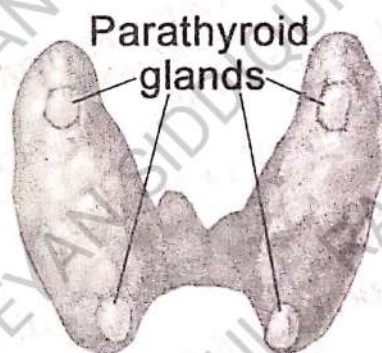
- Stunted growth.
- Mental retardation.
- Coarse facial features.
- Reproductive organs do not function normally.

ii. GOITER:

A goiter is an abnormal enlargement of thyroid gland. It occurs due to low intake of iodine in diet. It is usually painless. A large goiter can cause cough and make it difficult for one to swallow or breathe.

4. PARATHYROID GLAND:

These are two pairs of pea-sized glands, dorsally adjacent to the two lobes of thyroid gland in the neck. They release **Parathyroid hormone (PTH)** or **Parathormone** in response to low levels of calcium and regulate the serum calcium concentration.

**Functions:**

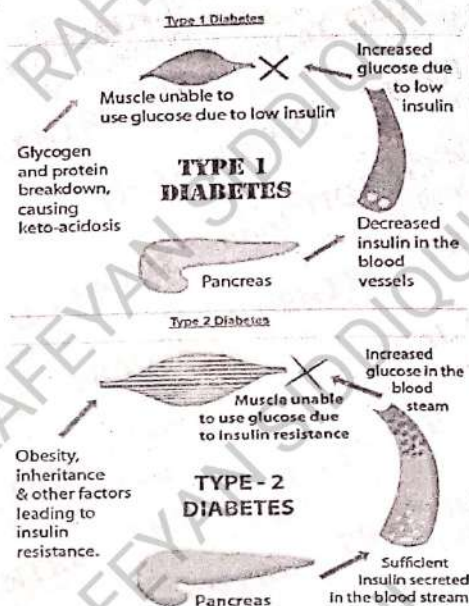
- It stimulates the reabsorption of calcium in the kidneys and leads to activation of Vitamin D needed for absorption of calcium in the intestine.
- It stimulates osteoclasts cells of bones to release calcium in the blood.

5. PANCREAS:

Pancreas is a heterocrine gland i.e., both exocrine and endocrine in function. Most of pancreatic cells are exocrine in function but there are patches of cells called "**Islets of Langerhans**" which have endocrine function. Islets of Langerhans have two types of cells; **alpha cells** which secrete Glucagon and **beta cells** which secrete Insulin. Both of these hormones have antagonistic effect and regulate blood glucose.

a) GLUCAGON:

It is secreted in response to low level of blood glucose. It activates a liver enzyme that breaks down glycogen-releasing glucose bringing glucose levels to normal in the



- Muscle wasting.
- Weak bones.
- Mood swings.
- Hypertension.

b) **ADDISON'S DISEASE:**

It occurs due to low secretion of cortisol and may also be a consequence of autoimmune process. Lack of cortisol results in drop of glucose and high susceptibility to any kind of stress due to insufficient energy supply. In this state, even a mild infection can cause death. Its symptoms include:

- Weakness.
- Weight loss.
- Low blood sugar.
- Reduced blood pressure.

ii. **MINERALOCORTICOID:**

They help to regulate mineral balance.

a) **ALDOSTERON:**

It increases the reabsorption of Na^+ and Cl^- ions by the kidney, thus regulating blood volume and blood pressure.

RAAS (Renin Angiotensin Aldosterone System):

- During low blood sodium level, the kidneys secrete an enzyme, renin which converts a plasma protein, angiotensinogen to angiotensin-I.
- Angiotensin-I is converted into Angiotensin in the lungs, which stimulates adrenal cortex to release aldosterone.
- Renin system affects blood in two ways. Firstly, angiotensin constricts the arteries raising blood pressure.
- Secondly aldosterone causes increase in sodium absorption, maintaining sodium levels. When sodium levels are high, secretion of ADH (anti-diuretic hormone) is stimulated which causes re-absorption of water and thus blood volume and blood pressure is maintained.

b) **ANDROGENS:**

Androgens are similar to testosterone (male hormone) and are produced in both the sexes. They affect bones and muscle development and metabolism.

In male, it is responsible for the development of secondary sexual characters such as growth of facial hairs, deepening of voice and increase in muscle bulk.

In females, one of its main purposes is to be converted into estrogen. However, its over-secretion may lead to masculinization. (e.g., appearance of beard).

ADRENAL MEDULLA:

The adrenal medulla works under the influence of sympathetic nervous system. It secretes epinephrine and nor-epinephrine during emergency situation (fight and flight conditions).

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i. Epinephrine (adrenaline):

It increase heartbeat, blood glucose, breathing rate and metabolic rate.

ii. Norepinephrine (nor-adrenaline):

It functions like epinephrine but its main function is to sustain blood pressure.

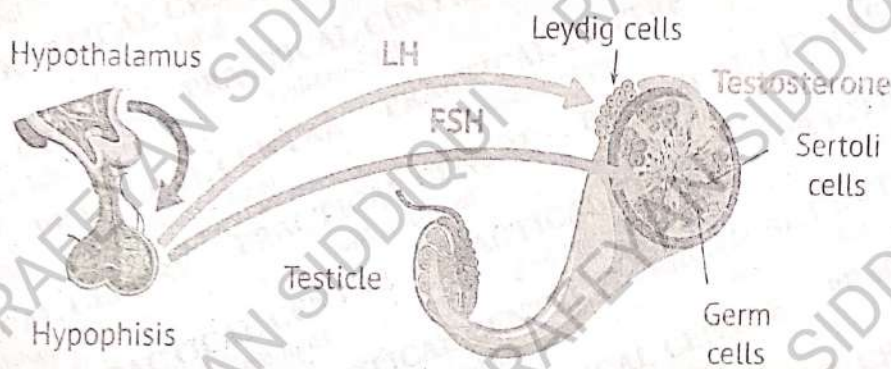
7. GONADS:

A gonad, sex gland or reproductive gland is a mixed gland that produces gametes as well as sex hormones in an organism. The male gonads are the **testes** and the female gonads are the **ovaries**.

i. TESTES:

Testes are male gonads which secretes male sex hormones. Follicle-stimulating hormone (FSH) released from anterior pituitary gland stimulates **Sertoli cells** of testes to facilitate sperm development.

LH released from anterior pituitary stimulates **Leydig cells** located in interstitial tissue of testes to release testosterone. It is produced early in the development of an embryo and ensures that the fetus develops into male.



EFFECT OF TESTOSTERONE:

Testosterone produces both anabolic and androgenic effects in males.

ANABOLIC EFFECTS:

It includes muscle mass, muscle strength, increased bone density, bone strength, linear growth and bone maturation.

ANDROGENIC EFFECTS:

It includes maturation of sex organs, formation of scrotum in fetus, deepening of voice and growth of facial hairs.

ii. OVARIES:

The ovaries are female gonads which produce both ovum and female sex hormones. Ovaries secrete two lipophilic hormones, oestrogen and progesterone.

ESTROGEN:

maintains female reproductive system and development of the secondary sexual characters in female such as breast development, fat distribution in hips, legs and breast, armpit and pubic hairs and menarch (start of menstrual cycle). Low levels of oestrogen cause mood swings, depression, headaches, irregular periods and sleep problems.

PROGESTERONE:

Progesterone is primarily involved in the preparation and maintenance of the uterus which supports the growth and development of an embryo. Thus, it helps in regulating menstruation and maintaining pregnancy.

8. PINEAL GLAND:

It is a tiny, cone-shaped body behind the third ventricle of the brain. It secretes **melatonin** which has various functions, most important of which is to help modulate sleep patterns. Its production is stimulated by darkness and inhibited by light.

OTHER ENDOCRINE TISSUES:**i. HORMONES OF DIGESTIVE TRACT:****➤ GASTRIN:**

It is a peptide hormone. It stimulates mucosal growth, gastric motility and secretion of HCl in stomach.

➤ SECRETIN:

It is a peptide hormone, produced by the wall of duodenum. It stimulates the release of bicarbonate and water from the pancreas to neutralize the acidity in the duodenum.

➤ CHOLECYSTOKININ:

It is also a peptide hormone, produced by the walls of duodenum. It stimulates the contraction of gall bladder and release of digestive enzymes from the pancreas.

ii. HORMONES OF KIDNEY:**➤ ANGIOTENSIN:**

It regulates blood pressure.

➤ ERYTHROPOIETIN:

It stimulates the production of RBCs in the bone marrow.

iii. HORMONES OF HEART:**➤ ATRIAL NATRIURETIC HORMONE:**

It stimulates the excretion of salt and water from the kidneys to reduce blood pressure.

iv. PROSTAGLANDINS:

The prostaglandins are a group of lipids, made at the sites of tissue damage or infection that are involved in dealing with injury and illness. They control processes such as

inflammation, blood flow, formation of blood clots and the induction of labour.

v. HORMONES OF PLACENTA:

- **HUMAN CHORIONIC GONADOTROPIN (HCG):** It is also known as the pregnancy hormone, because it supports the development of the foetus.
- **PROGESTERONE:** It helps in regulating menstruation and maintaining pregnancy.

FEEDBACK MECHANISM:

Chemical co-ordination is regulated by feedback mechanism. Feedback mechanism is the check and balance mechanism, through which homeostasis is carried out in the body.

TYPES OF FEEDBACK MECHANISMS:

There are two mechanisms to maintain blood hormonal levels i.e., negative and positive feedbacks.

vi. NEGATIVE FEEDBACK: It refers to the opposite effect produced in relation to any change in the body fluids. e.g.: eating rich sugary food causes raised glucose level in the blood. It triggers the mechanism, which decreases the blood glucose back to normal limit.

vii. POSITIVE FEEDBACK:

It refers to the series of similar effects produced in relation to the change in the body fluids. E.g.: during childbirth, oxytocin is released from posterior lobe of pituitary gland. It stimulates the contraction of uterus. The uterine contractions cause further release of oxytocin. Similarly, during breastfeeding, oxytocin is released which stimulates the release of milk. The milk ejection stimulates further release of oxytocin.